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Inter-polymer complex microspheres of chitosan and cellulose acetate phthalate for oral delivery of 5-fluorouracil

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Abstract Inter-polymer complexes (IPCs) of chitosan (CS) and cellulose acetate phthalate (CAP) have been prepared to develop spherical microspheres by a novel emulsion-solvent evaporation technique. The microspheres were used for the oral delivery of 5-fluorouracil (5-FU), an antimetabolite and antineoplastic agent, whose release time was extended up to 12 h. Formulations were prepared by varying the concentrations of CS, CAP and 5-FU. FTIR confirmed the formation of IPC, indicating no chemical interactions of 5-FU with the polymer matrix. Scanning electron microscopy suggested spherical shape of the microspheres with smooth surfaces. Average particle size measured by optical microscopy varied between 2.7 and 5.5 μm. Differential scanning calorimetry showed amorphous dispersion of 5-FU particles into the IPC matrix. Encapsulation efficiency as estimated by UV was dependent on polymer composition with the highest value of 96 %. Water uptake by the IPC microspheres was higher at higher concentration of CS in the matrix. In vitro drug release performed in pH 1.2 and pH 7.4 buffer media showed a dependence on compositions of CS, CAP and drug loading. Molar mass between cross-links (M_c) and cross-link density (d_x) values of the polymer matrix calculated from swelling data indicated the formation of a dense matrix between CS and CAP; the matrix was able to control the release of 5-FU. The in vitro release data have been fitted to empirical equations to understand the nature of drug release mechanism.

This paper is dedicated to the fond memory of late Professor Shalin Thakkar.

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Introduction

Novel drug delivery systems (NDDS) based on biodegradable and natural polymers have been the attractive biomaterials in pharmaceutical sciences, because such formulations extend the release time of short-lived drugs [1]. These systems have been widely explored with a view to release the drugs at a controlled rate by maintaining constant and required plasma concentrations over an extended period of time causing minimum side effects. Even though significant progress has been made over the past decades in developing NDDS as micro/nanoparticulate systems, still there is a continuous spur of research activity in developing such formulations for delivering drugs for treatment of various disease problems [2–5]. It is always desirable to administer the formulated drugs in a specific and controlled manner for precisely matching the physiological needs at appropriate times (temporal modulation) and/or at appropriate site (site-specific delivery) [6]. To accomplish these goals, various strategies have been developed using the natural and/or synthetic polymers as controlled release (CR) dosage forms [7–9]. Of the many routes of drug administration, oral route has been the most widely preferred approach due to its several inherent advantages such as ease of administration and patient compliance [10]. Therefore, the demand for developing CR oral formulations continues to be widely accepted by the pharmaceutical industries [11].

In recent years, there is a growing interest in developing inter-polymer complexes (IPCs) that are generally formed by the interaction of two oppositely charged polymers, resulting in self-assembly or spontaneous association of polymer chains due to the formation of strong and reversible electrostatic links [12–14]. The formation of such IPCs with non-permanent structures possibly avoids the use of toxic covalent bonded chemical cross-linkers, thereby rendering their safe utility in human use. Such networks are well tolerated and are needed due to their biocompatibility and sensitivity to changes in environmental conditions [15, 16]. While there is a choice of many polymers to develop IPCs, chitosan (CS), a natural polysaccharide with its cationic nature, has been widely used because of its biocompatibility and biodegradability [17–19]. Moreover, CS is known to possess mucoadhesive properties due to inter-molecular attractive forces formed by electrostatic interactions between the positively charged CS and the negatively charged mucosal surfaces [20, 21]. The IPCs formed from CS with the selected natural and/or synthetic polyanions have therefore been widely used as excipients in modifying the release rates of drugs such as in matrix type tablets, beads, hydrogels, microparticles or nanoparticles through different routes of administration [12, 22–28].

Cellulose acetate phthalate (CAP), a partially *o*-acetylated and *o*-phthalylated cellulose, has been used as an enteric film coating material or as a matrix binder for tablets and capsules [29]. This type of coating resists the prolonged contact with strongly acidic gastric fluid, but dissolves in mild acidic or alkaline environment of

the intestine. CAP is commonly applied in solid dosage forms either by coating from organic or aqueous solvent systems or by direct compression. It is also used in combination with other coating agents such as ethyl cellulose in CR preparations. For these reasons, CAP is chosen to develop IPCs in combination with CS.

The model drug chosen in this work is 5-fluorouracil (5-fluoro-2, 4-pyrimidinedione), (5-FU), an antimetabolite of the pyrimidine analogue type with a broad spectrum of activity against solid tumors (of the gastrointestinal tract, pancreas, ovary, liver, brain, breast, etc.). Due to its structure, 5-FU interferes with the nucleoside metabolism and thus can be incorporated into RNA and DNA, leading to cytotoxicity and cell death [30]. The biological half-life of 5-FU is $\approx$10–20 min. However, due to its high rate of metabolism in the body, the maintenance of its therapeutic serum concentration requires the continuous administration of high doses, but concentrations above a certain limit would produce severe toxic effects. To increase the therapeutic efficacy of 5-FU, several CR systems have been developed in the literature using biopolymers [4–6, 31–33]. But their encapsulation efficiencies were quite low, and to further increase the effectiveness of such formulations, we have chosen to develop a novel oral delivery CR system containing 5-FU to increase its therapeutic efficacy with reduced toxicity using biodegradable CS/CAP-based IPCs.

The cationic nature of chitosan (highly biocompatible and biodegradable) has been exploited to develop inter-polymer complexes with anionic polymers of both natural and synthetic origins [17]. CAP is selected due to its anionic nature and is widely used as an enteric coating material in many pharmaceutical oral dosage forms [34]. The phthalate group of CAP is amenable to the formation of IPC with the cationic chitosan. By this approach, toxic cross-linking agents can be avoided, since only weak hydrogen bonds and electrostatic interactions are responsible to form the IPCs. The CR formulations have been used to investigate the in vitro release profiles of 5-FU in acidic (pH 1.2) and alkaline (pH 7.4) phosphate buffer media. The in vitro release data have been analyzed using the empirical equations to understand the release profiles of 5-FU.

Materials and methods

5-Fluorouracil was purchased from Hi Media Chemicals, Mumbai, India. Chitosan [medium molecular weight (Brookfield viscosity of 200,000 cps) and degree of deacetylation of 75–85 %] was procured from Aldrich Chemical Company, Milwaukee, WI, USA. Cellulose acetate phthalate (CAP) was purchased from Yarrow Chemical Products, Mumbai, India; Span-80, *n*-hexane, acetone, light liquid paraffin oil and glacial acetic acid were all procured from s.d. fine chemicals, Mumbai, India; potassium dihydrogen orthophosphate and sodium hydroxide were purchased from Hi Media Chemicals, Mumbai, India. All other chemicals were used without further purification.

Preparation of IPC microspheres

The IPC microspheres of CS and CAP containing 5-FU were prepared by emulsion-solvent evaporation method as displayed schematically in Fig. 1. In our novel approach, a specified amount of 5-FU was accurately weighed and placed in a 50 mL beaker containing 20 mL of 1 % aqueous acetic acid solution. The beaker was placed in a sonicator for 10 min to completely dissolve 5-FU to which a specified amount of CS was added slowly under stirring on a magnetic stirrer for overnight until a homogeneous mixture was obtained. This was further emulsified into 100 mL of light liquid paraffin oil containing 1 % Span-80 at 800 rpm using an over-head stirrer (Remi Equipments, Mumbai, India) to obtain the w/o emulsion.

Separately, a specified amount of CAP was dissolved in 10 mL of acetone: water mixture (97:3) [36] and stirred for 15 min to obtain a homogeneous solution. This solution was added drop-wise to the above prepared w/o emulsion and stirred for 2 h to completely evaporate acetone. The IPC microspheres were formed due to the interaction between positive charges of CS and negatively charged sites of CAP. The microspheres thus formed have the net positive charges and were allowed to settle down. The upper oily layer was decanted; the microspheres were washed 2–3 times with *n*–hexane and dried in an incubator at 40 °C for overnight. Dried microspheres were placed in a vacuum desiccator before further evaluations. Totally, eight formulations were prepared by taking different amounts of CS, CAP and 5-FU as per the formulation codes assigned from F1 to F8 and these are presented in Table 1. A schematic representation of the formation of IPC between CS and CAP is displayed in Fig. 2.

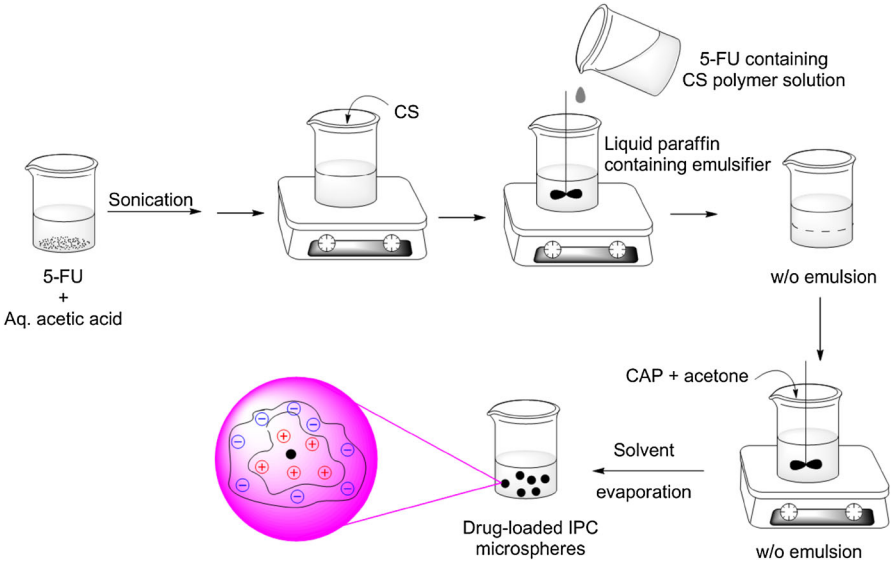

Fig. 1 Preparation method to produce IPC microspheres

Table 1 Formulation details along with results of % yield, % encapsulation efficiency, average particle size, % equilibrium swelling, molar mass between cross-links and cross-link density

Formulation codes	Chitosan (g)	CAP (g)	Drug loading (g)	% EE[a]	Average[a] particle size (μm)	% Equilibrium[a] swelling	M_c	$d_x \times 10^4$
F1	0.3	0.3	0.1	89.5 ± 1.5	2.7 ± 0.19	70 ± 0.86	8,576	2.04
F2	0.3	0.3	0.2	32.5 ± 1.0	2.9 ± 0.20	108 ± 0.76	7,161	2.50
F3	0.3	0.5	0.1	90.5 ± 2.1	2.8 ± 0.17	42 ± 0.57	5,882	2.98
F4	0.3	0.5	0.2	35 ± 2.5	2.7 ± 0.24	56 ± 1.04	5,240	3.34
F5	0.5	0.3	0.1	91.6 ± 2.3	2.8 ± 0.26	143 ± 0.58	4,510	3.89
F6	0.5	0.3	0.2	32.5 ± 1.2	4.1 ± 0.18	171 ± 1.10	3,361	5.21
F7	0.5	0.5	0.1	95.6 ± 1.1	5.1 ± 0.16	100 ± 1.0	3,300	5.31
F8	0.5	0.5	0.2	38.8 ± 1.8	5.5 ± 0.29	125 ± 1.15	2,800	6.26

% EE % encapsulation efficiency

[a] Presented data are mean ± SD, $n = 3$

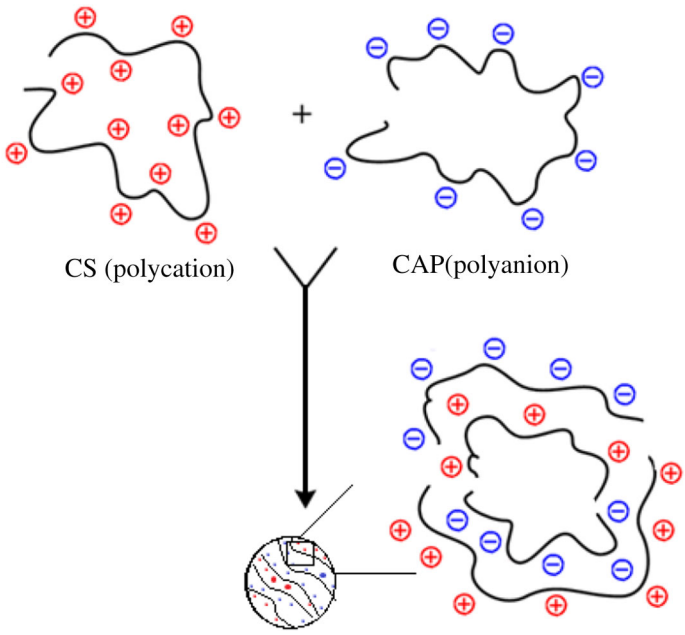

Fig. 2 Schematic representation of the formation of inter-polymer complex between CS and CAP

Drug content

Drug-loaded microspheres were accurately weighed (40 mg), ground to fine powder using mortar and pestle, drug was extracted using distilled water and taken in 50 mL volumetric flask and its volume was made up to mark. The flask was kept in an orbital shaker incubator maintained at 37 °C for overnight at 100 rpm rotation

speed. Then the solution was filtered through Whitman filter paper and the clear supernatant solution obtained was analyzed by UV spectrophotometer keeping the fixed λ_{max} value of 265 nm (LabIndia, UV 3000, Mumbai, India). Analysis was done in triplicate for each sample, but the average values were considered in data analysis and graphical display.

The % encapsulation efficiency (EE) was calculated as:

$$\% \text{ Encapsulation efficiency} = \left(\frac{\text{Drug loading}}{\text{Theoretical drug loading}}\right) \times 100 \qquad (1)$$

These data for various formulations are included in Table 1.

Particle size measurements

Determination of average particle size of 5-FU-loaded microspheres was carried out using a trinocular microscope (LABOMED Inc., USA). A small quantity of microspheres was spread onto a clean glass slide on which a drop of light liquid paraffin oil was placed to separate the particles and covered with a cover slip. Sizes of hundred particles were measured to determine the average particle size for each formulation by triplicate measurements. Standard deviations were calculated by the method of least squares at 95 % confidence limit and these data are also included in Table 1.

Scanning electron microscopic (SEM) studies

SEM was used to determine the surface morphology of IPC microspheres. The SEM Quanta 200 scanning microscope available at Indian Institute of Science, Bangalore, India was used. Dry microspheres were placed on an electron microscope brass stub and coated with gold in an ion sputter. SEM micrographs of the microspheres were taken by random scanning of the stub.

Fourier transform infrared (FTIR) spectral measurements

FTIR spectral data were taken on a Bruker (Alpha series, Germany) instrument to confirm the IPC structure formation as shown in Fig. 3 as well as to find the chemical stability of 5-FU in the formulated microspheres. FTIR spectra of placebo microspheres, drug-loaded microspheres and pristine drug were obtained by crushing the microspheres with KBr to get pellets under a pressure of 600 kg/cm^2. Spectral scanning was done in the range between 4,000 and 500 cm^{-1}.

Differential scanning calorimetric (DSC) studies

DSC (TA Instruments, USA) was performed on drug-loaded microspheres, placebo microspheres and pristine drug. Samples were heated from 25° to 400 °C at the heating rate of 10 °C/min in a nitrogen atmosphere (flow rate 10 mL/min).

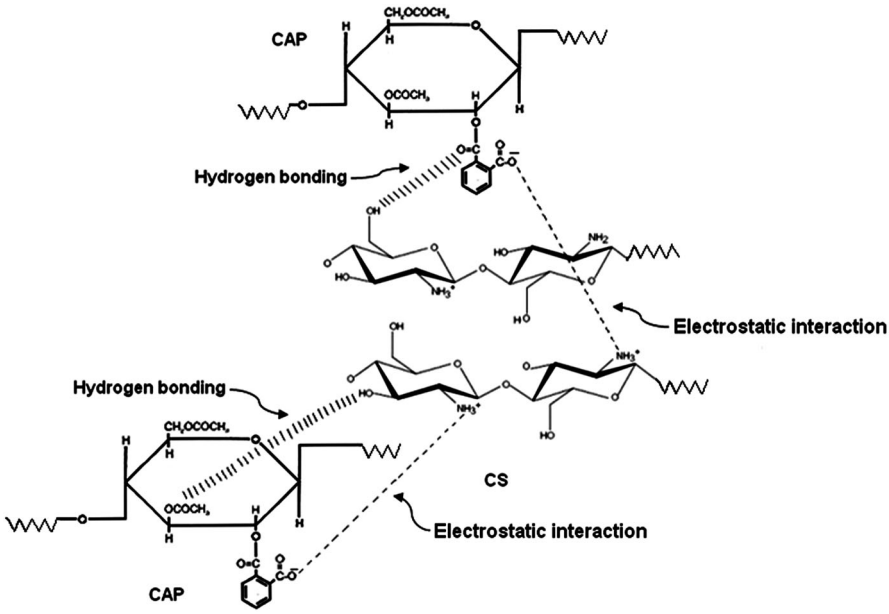

Fig. 3 Interaction model of inter-polymer complex between CS and CAP

Equilibrium swelling studies

Equilibrium swelling measured in terms of water uptake by the microspheres was determined gravimetrically by measuring the extent of matrix swelling in aqueous media. To ensure complete equilibration, the samples were swollen for 24 h, excess surface-adhered liquid droplets were removed by blotting with soft tissue papers and weighed accurately to ±0.01 mg using electronic microbalance (Sartorius, BS 224 S, Gottingen, Germany). The microspheres were then dried in an oven at 60 °C for 5 h until no change in the mass of the dried sample was observed. Experiments were performed in triplicate for each sample, but the average values were considered in data analysis. Standard deviations in % swelling were calculated by the method of least squares at 95 % confidence limit as:

$$\% \text{ Swelling} = \left(\frac{\text{Weight of swollen microspheres} - \text{Weight of dry microspheres}}{\text{Weight of dry microspheres}}\right) \times 100 \tag{2}$$

In vitro dissolution studies

Drug release from the IPC microspheres with different 5-FU loadings and polymer compositions were investigated first in pH 1.2 acidic buffer solution, initially for 2 h, and then in pH 7.4 phosphate buffer solution until the completion of

dissolution. The experiments were performed using USP XXIII (Electrolab, Model TDT 08L, India) equipped with eight baskets at the stirring speed of 100 rpm. A weighed quantity of each sample equivalent to 50 mg of 5-FU was placed in 900 mL of dissolution medium maintained at 37 °C. A 5 mL of the sample was withdrawn through a hypodermic syringe fitted with a 0.4 μm Millipore filter from the dissolution media at suitable time intervals and the same amount was replaced with fresh buffer solution to maintain the sink conditions. The 5-FU concentration was then determined using UV spectrophotometer at the λ_{max} of 265 nm. These studies were performed in triplicate for each sample, but the average values were considered in data analysis by calculating the standard deviations by the method of least squares at 95 % confidence limit.

Model fitting of in vitro release data

Kinetics of 5-FU release from all the formulations was investigated by fitting the in vitro release data to Zero order, First order, Higuchi square root, Hixson–Crowell cube root as well as Ritger–Peppas equations using the Microsoft excel 97 to compute regression coefficients, r^2, approaching close to the unity that were considered as the best fit model for the system under consideration.

Statistical analysis

Results were analyzed and expressed as mean ± SD. Effect of compositions of CS and CAP on encapsulation efficiency, particle size, % swelling and drug concentration on in vitro release from the IPC microspheres were all statistically analyzed using the one-way ANOVA method followed by Tukey's multiple comparison test. The differences were considered significant at the level of $p < 0.0001$, where p is the probability value. GraphPad Prism 5 software was used in all the statistical analyses.

Results and discussion

Preparation and characterization of microspheres

In this work, 5-FU-loaded IPC microspheres of CS and CAP were prepared by a novel modified emulsion-solvent evaporation technique. The formation of IPC occurs by in situ precipitation of CS droplets in acetone followed by interaction with CAP. The IPC microspheres are formed due to the interaction between the positive charges of CS and the negatively charged sites of CAP. Thus, formed microspheres possess the net positive charges and the negatively charged CAP will be at the partial outer layer to control further release of 5-FU. The net partial positive charge of IPC get stabilized through electrostatic interactions, but much weaker hydrogen bond interactions may not be responsible to control the release of 5-FU as can be seen from the interaction model displayed in Fig. 3.

From the results of % encapsulation efficiency (EE) shown in Table 1, the EE values vary from 33 to 96 %, depending upon the composition of CS and CAP in the IPC matrix. For instance, formulations F1 (CS = 1.5 %, CAP = 0.3 g and 5-FU = 0.1 g) and F2 (CS = 1.5 %, CAP = 0.3 g and 5-FU = 0.2 g) have the EE values of 90 % and 33 %, whereas formulations F3 (CS = 1.5 %, CAP = 0.5 g and 5-FU = 0.1 g) and F4 (CS = 1.5 %, CAP = 0.5 g and 5-FU = 0.2 g) have the EE values of 91 and 35 %, respectively. This indicates the dependence of EE on polymer composition and drug loading. Similarly, formulations F7 (CS = 2.5 %, CAP = 0.5 g and 5-FU = 0.1 g) and F8 (CS = 2.5 %, CAP = 0.5 g and 5-FU = 0.2 g) exhibit the EE values of 96 and 39 %, respectively, whereas formulations F5 (CS = 2.5 %, CAP = 0.3 g and 5-FU = 0.1 g) and F6 (CS = 2.5 %, CAP = 0.3 g and 5-FU = 0.2 g) have shown, respectively, the EE values of 92 and 33 %.

The EE results are statistically significant giving a probability value of $p < 0.0001$ due to the formation of a more dense IPC structure as evidenced by the higher value of $pK_a = 6.5$ for CS compared to the pK_b value of 5.0 for CAP, signifying the net partial positive surface charge of the IPC. Such a matrix would be able to hold a large amount of 5-FU, which explains higher value of EE of 96 % as displayed with F1, F3, F5 and F7. Statistical analysis by ANOVA showed significant effect of CS on % EE ($p < 0.0001$) values as well as drug loading also exerts an effect on EE data. However, lower values of EE are observed for F2, F4, F6 and F8 due to increase in concentration of drug in the formulation.

The average particle size of microspheres showed an effect on IPC composition as well as drug loading, which increased from 1.7 to 5.5 μm for 1.5–2.5 % for CS and 300–500 mg for CAP containing IPC. Particle size showed a dependence on the amount of CS as well as CAP of the IPC matrix. Notice that formulations with higher concentrations of CS (2.5 %) and CAP (500 mg) exhibited higher particle size i.e., F5 and F6 showed higher particle size of 2.8 and 64 μm, respectively, than formulations F11 and F24 that showed smaller particle sizes of 2.7 and 8.9 μm, respectively, with values of $p < 0.0001$. Similarly, formulations F7 and F8 showed higher particle sizes of 5 and 5.5 μm, respectively, than those of F5 and F6 formulations, with the value of $p < 0.0001$. Viscosity of the medium has influenced the particle size i.e., viscosity increases with increasing polymer concentration, leading to the formation of larger polymer droplets, thereby increasing the size of the microspheres. SEM confirmed the spherical nature of particles with smooth surfaces as shown in Fig. 4 for group of particles as well as single particles.

Fourier transform infrared spectral (FTIR) studies

Formation of IPC was confirmed by FTIR as displayed in Fig. 5 for plain CS, CAP and placebo microspheres. In case of CS, a broad band at 3,422 cm^{-1} indicates –N–S stretching vibrations of the amine group, whereas for CAP, a sharp band at 1,678 cm^{-1} represents –C=O stretching of phthalate group of CAP. In placebo microspheres, –N–H stretching band of CS and $–COO^-$ stretching band of CAP have diminished, indicating the formation of IPC structure between CS and CAP

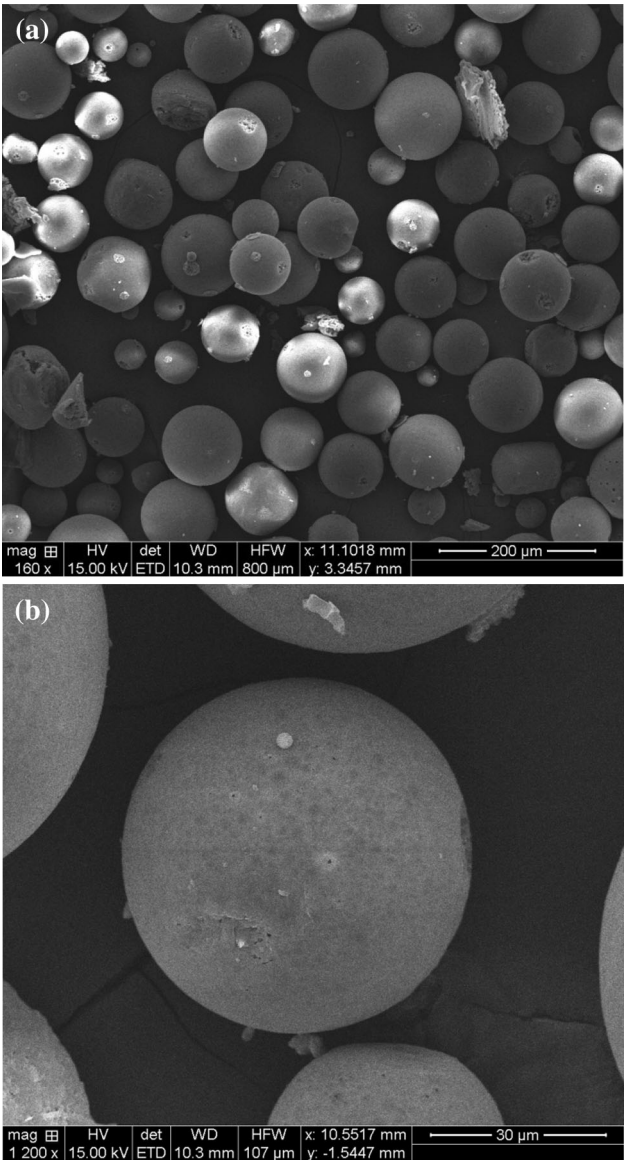

Fig. 4 SEM micrographs of **a** group of particles (F4) and **b** single particles (F4)

due to the interaction of $–NH_2$ of CS with the phthalate group of CAP, leading to the net positive charges as explained before.

Chemical stability of 5-FU in IPC microspheres was also confirmed by FTIR as shown by the spectra of drug-loaded microspheres and 5-FU displayed in Fig. 6, wherein 5-FU exhibits a band at 3,134 cm^{-1}, indicating N–H stretching vibrations. However, the bands at 1,724 and 1,669 cm^{-1} indicate S=O stretching vibrations. A

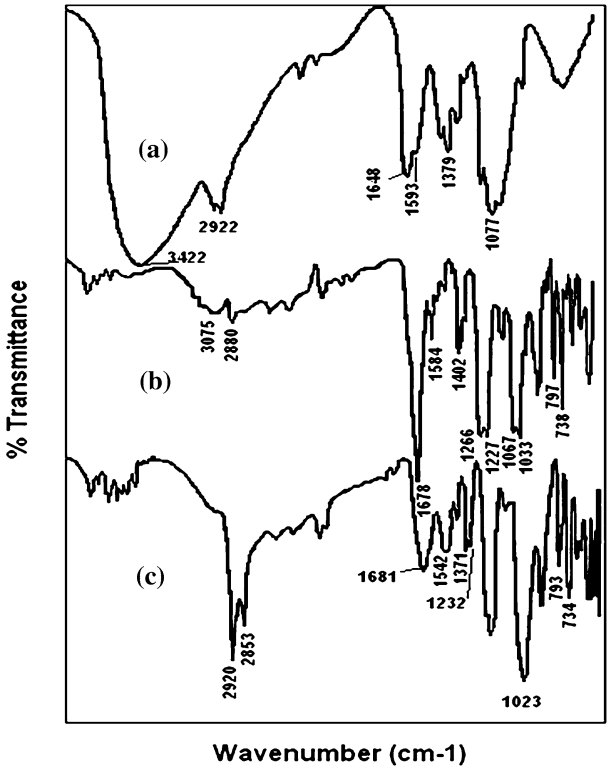

Fig. 5 FTIR spectra of *(a)* CS, *(b)* CAP and *(c)* placebo microspheres

band at 1,247 cm^{-1} represents C–H in-plane deformation, but a band at 815 cm^{-1} indicates the C–H out-of-plane deformation. In drug-loaded formulations, the above-mentioned characteristic bands of 5-FU are also present at 3,125 cm^{-1} for N–H stretching vibrations as well as at 1,716 and 1,654 cm^{-1} for C=O stretching vibrations. The bands at 1,239 cm^{-1} for C–H in-plane deformation and 805 cm^{-1} for C–H out-of-plane deformation, respectively, indicate the chemical stability of 5-FU in drug-loaded microspheres.

Differential scanning calorimetric (DSC) studies

DSC was performed to investigate the crystalline nature of 5-FU in the formulated microspheres. DSC thermograms of (a) placebo microspheres, (b) 5-FU-loaded microspheres and (c) pristine 5-FU displayed in Fig. 7 shows two peaks for placebo microspheres of which endothermic peak observed at 100.9 °C is attributed to the loss of volatile components (acetone and moisture) or possible chain relaxation, while the exothermic peak is observed at 241.9 °C due to the degradation of polymer matrix. The electrostatic interactions between CS and CAP polymers have caused the loss of H bonding [35]. Thermogram of 5-FU showed a sharp peak at

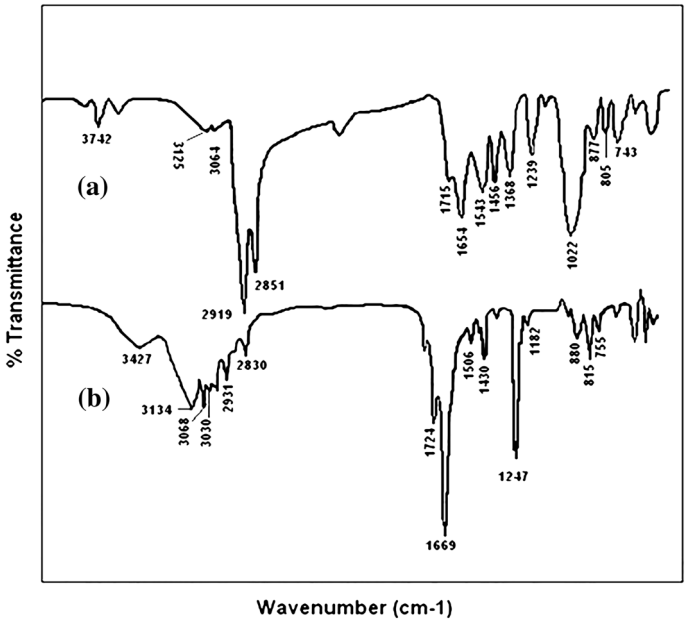

Fig. 6 FTIR spectra of *(a)* 5-FU-loaded microspheres and *(b)* pure 5-FU

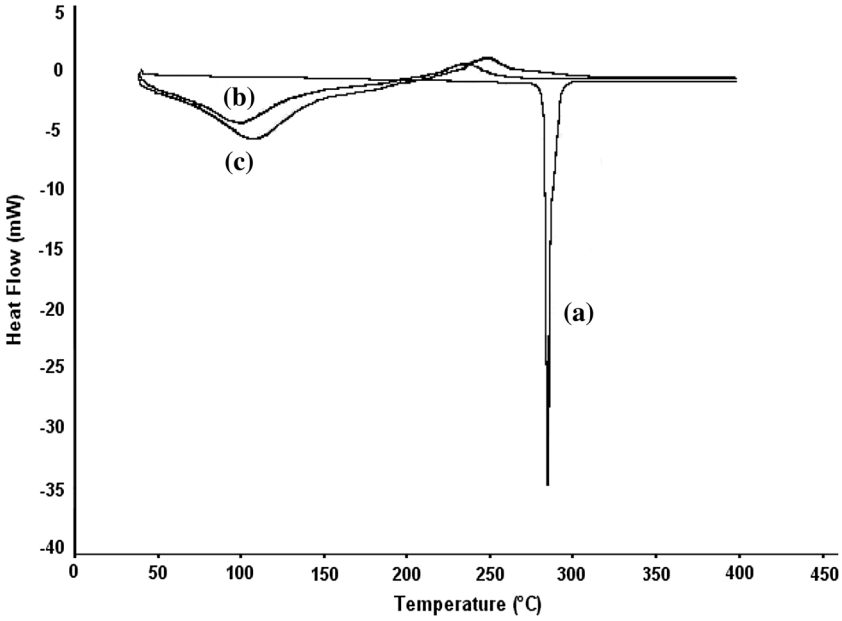

Fig. 7 DSC thermograms of *(a)* pure 5-FU, *(b)* placebo microspheres and *(c)* 5-FS-loaded microspheres

284.41 °C, indicating the melting point [36] of 5-FU. In case of 5-FU-loaded microspheres, two peaks are observed, one referring to endothermic peak at 108.81 °C, indicating the loss of volatile components (acetone and moisture) or chain relaxation and another at 234.81 °C due to the degradation of polymer matrix; but no peak corresponding to 5-FU is observed, indicating the amorphous dispersion of 5-FU particles in the IPC matrix.

Equilibrium swelling studies

The % equilibrium of 5-FU-loaded IPC microspheres indicates that with increasing CAP concentration of the formulation, the % equilibrium swelling is decreased due to hydrophobic nature of CAP, leading to lesser sorption of water molecules from the IPC microspheres. For instance, formulations F1 and F2 that contained 0.3 g of CAP showed % equilibrium swelling of 70 and 108, respectively, compared to lower values of 42 and 56 % for F3 and F4, which contained 0.5 g of CAP. Thus, as the CAP content of the IPC is increased from 0.3 to 0.5 g, a drastic reduction in % equilibrium swelling is observed. Statistical analysis also showed a significant difference of $p < 0.0001$.

With increasing concentration of CS, % equilibrium swelling has increased as shown in case of F5 and F6, which contained 0.5 g of CS. These displayed % equilibrium swelling values of 143 and 171, respectively, compared to the lower values of 70 and 108 for formulations F1 and F2 that contained 0.3 g of CS. Thus, as the amount of CS increases, there is an increase in water retention capacity of the matrix due to the formation of a dense gel structure. The ANOVA analysis of CS content on % equilibrium swelling of IPC microspheres also displayed a significant difference ($p < 0.0001$).

Evaluation of network parameters

Drug release depends on the density, d_x of matrix as well as molar mass, M_c between junction points of the network. It is therefore, useful to understand the IPC network properties. To understand the rigidity of IPC network, we have calculated d_x and M_c values using the Flory–Rehner theory based on equilibrium swelling data of the matrix [37]. If a polymer is placed in a solvent, it swells until the elastic forces due to stretching of the polymer chain segments balance the osmotic forces. Such elastic retractive forces are inversely proportional to molar mass, M_c of the polymer between points of cross-linking. If M_c values between two junction points in a polymer network are high, then the matrix becomes rigid, thereby exhibiting limited swelling. On the other hand, if M_c is large, the network becomes more elastic and swells rapidly in the presence of liquid. Flory–Rehner equation [37] in the following form was used to calculate M_c.

$$M_c = -\rho_p V_s \varphi^{1/3} \left[\ln(1-\varphi) + \varphi + \chi\varphi^2\right]^{-1} \quad (3)$$

In the above equation, volume fraction, φ of the polymer in its swollen state was calculated using:

$$\varphi = \left[1 + \frac{\rho_p}{\rho_s}\left(\frac{M_a}{M_b}\right) - \frac{\rho_p}{\rho_s}\right]^{-1} \tag{4}$$

Here, ρ_p and ρ_s are densities of polymer and solvent, respectively; M_b and M_a are, respectively, the masses of polymer before and after swelling, and V_s is molar volume of the solvent. The interaction parameter, χ that represents the compatibility between solvent–polymer pair was calculated from the Bristow and Watson equation [38]:

$$\chi = \beta + \left(\frac{V_s}{RT}\right)\left(\delta_s - \delta_p\right)^2 \tag{5}$$

Here, β is a lattice constant, whose value is taken as 0.34, V_s is molar volume of the solvent, R is the molar universal gas constant, and T is temperature in Kelvin. The symbols δ_s and δ_p are solubility parameters of solvent and polymer, respectively.

The cross-link density (d_x) of the polymer matrix was calculated as [39]:

$$d_x = \left(\frac{1}{vM_c}\right) \tag{6}$$

Here, v is specific volume of the polymer. The calculated values of M_c and d_x given in Table 1 indicate widely varying trends. For instance, with formulations F1–F4 that contained 0.3 g of CS and with increasing concentration of CAP (0.3 and 0.5 g), the M_c values decrease considerably from 8,576 to 540, but d_x values increase from 2.04×10^{-4} to 3.34×10^{-4}. A similar, but decreasing trend in M_c values is observed from 4,510 to 2,800 for formulations F5 to F8 that contained 0.5 g CS with varying concentration of CAP (0.3 and 0.5 g), but the d_x values increase considerably from 3.89×10^{-4} to 6.26×10^{-4}. Thus, the results of drug loading and EE values affect the morphological parameters. Overall, a good correlation is observed between the network parameter values and size of the microspheres.

In vitro drug release studies

In vitro 5-FU release from the IPC microspheres depends on the extent of drug loading. Release profiles of F1, F2, F5 and F6 formulations shown in Fig. 8 indicate that as the drug loading increases, the % cumulative release also increases [40]. However, noticeable changes in the % cumulative release curve are seen with the pristine 5-FU, which showed a quick release within 1–2 h. On the other hand, formulation F2 loaded with 0.2 g of 5-FU showed higher % cumulative release than F1 that contained 0.1 g of 5-FU. Similarly, formulations F4, F6 and F8 containing 0.2 g of 5-FU exhibit higher % cumulative release than those of F3, F5 and F7, which contained 0.1 g of 5-FU. Thus, % cumulative release data are higher for formulations containing higher amount of drug and vice versa. The ANOVA analyses of every hour release of 5-FU showed a significant difference ($p < 0.0001$). During the first 2 h of release study, dissolution was performed in

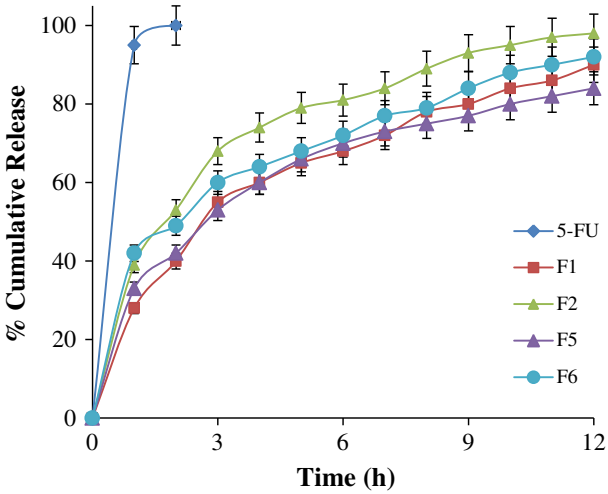

Fig. 8 Effect of concentration of 5-FU loading on in vitro release profiles for formulations F1, F2, F5 and F6

pH 1.2 buffer media, wherein burst release is observed and the release was extended up to 12 h.

Drug release showed a dependence on the amount of CS as shown in Fig. 9 with 5-FU and those of F1, F5, F2 and F6 formulations. An increase in the concentration of CS slowed the release of 5-FU compared to the matrix containing lower amount of CS [41]. For instance, formulation F1 (1.5 %, 300 mg of CS) has a higher % cumulative release than F5 (2.5 %, 500 mg of CS). Moreover, F2, F3 and F4 (1.5 %, 300 mg of CS) exhibit higher % cumulative release than those of F6, F7 and F8 (2.5 %, 500 mg of CS). Statistical analyses of every hour release data of the formulations indicate a significant difference ($p < 0.0001$). This is due to the fact that as the amount of CS in the matrix is increased, it forms a denser matrix, making it difficult for 5-FU particles to diffuse through the viscous gel layer of the matrix.

The % cumulative release also showed a dependence on the amount of CAP as shown in Fig. 10 for the release profiles of 5-FU in F5, F7, F6 and F8 formulations. Formulations F5 and F6 (0.3 g of CAP) showed higher % cumulative release than those of F7 and F8 (0.5 g of CAP). Similar results are observed for the remaining formulations. ANOVA analyses of every hour release of the formulations showed a significant difference ($p < 0.0001$). This is due to the formation of a more stable IPC structure at higher concentration of CAP, thereby retarding the release of 5-FU from the matrix [42]. In all the formulations (F1 to F8), nearly 70–80 % of total cumulative release of 5-FU occurred within 12 h, suggesting that the developed formulations for 5-FU, follows a controlled release pattern.

Model fittings

All the in vitro release data were fitted to empirical equations using the Microsoft Excel 97 to evaluate regression coefficients, r^2 for all the empirical equations

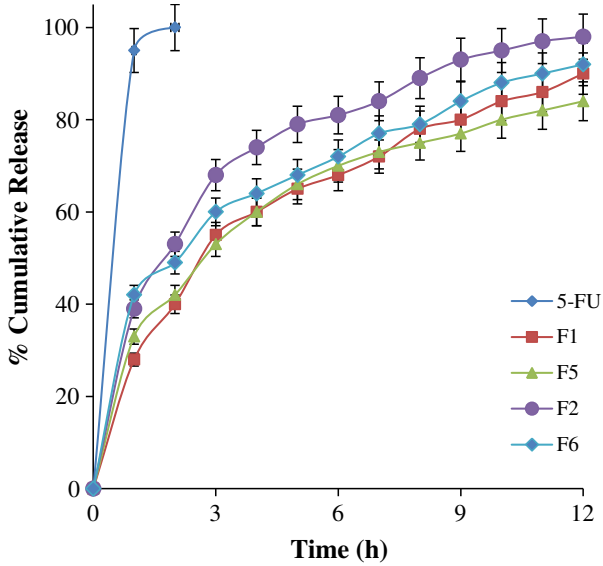

Fig. 9 Effect of concentration of CS on in vitro release profiles for formulations F1, F5, F2 and F6

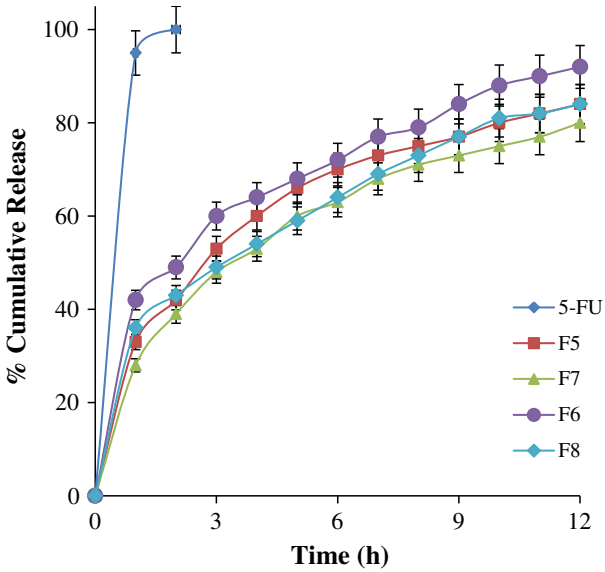

Fig. 10 Effect of concentration of CAP on in vitro release profiles for formulations F5, F7, F6 and F8

presented in Table 2. The value of r^2 of 0.984 was observed for Ritger–Peppas [43] equation in case of F1 with the exponent n value of 0.56. Similarly, r^2 values of 0.991, 0.990, 0.985 and 0.997 were observed for Ritger–Peppas equation for F2, F3, F4 and F7, respectively, with the n values of 0.49, 0.58, 0.51 and 0.47, suggesting

Table 2 Correlation coefficients, r^2, computed from different empirical equations

Formulations	Zero order	First order	Higuchi	Hixson–Crowell	Ritger–Peppas[a]	
	r^2	r^2	r^2	r^2	N	r^2
F1	0.967	0.977	0.977	0.959	0.56	0.984
F2	0.971	0.988	0.987	0.968	0.49	0.991
F3	0.958	0.979	0.988	0.960	0.58	0.990
F4	0.945	0.969	0.978	0.943	0.51	0.985
F5	0.993	0.995	0.992	0.994	0.45	0.990
F6	0.972	0.976	0.981	0.975	0.31	0.977
F7	0.974	0.988	0.996	0.975	0.47	0.997
F8	0.994	0.999	0.998	0.995	0.29	0.993

r^2 is correlation coefficient calculated by the method of least squares at 95 % confidence limit

[a] Estimated using Ritger and Peppas equation [4]: $\log\left(\frac{M_t}{M_\infty}\right) = \log k + n \log t$ where k is release kinetics constant, M_t and M_∞ are the cumulative releases at time, t and at longer time

that 5-FU release follows non-Fickian behavior. However, r^2 values of 0.995 and 0.999 were observed for First-order equation in case of F5 and F8 formulations and r^2 of 0.981 was observed for Higuchi equation for F6. It may be noted that varying trends in statistical analysis by all the empirical equations are due to the limits imposed by the statistical principles.

Conclusions

Using a novel approach, inter-polymer complex microspheres of CS and CAP were prepared by adapting an emulsion-solvent evaporation method. These formulations were then used to investigate the CR of 5-FU. The IPC microspheres were characterized by a variety of techniques and when subjected to in vitro release of 5-FU in both acidic and alkaline buffer media these showed maximum encapsulation efficiency up to 96 %. Spherically shaped microspheres have the smooth surfaces as confirmed by SEM. Average particle size of microspheres varied between 2.7 and 5.5 μm and 5-FU release was dependent on the amount of drug loading as well as CS and CAP compositions. It was found that IPC controlled the overall swelling of the matrix to facilitate the CR of 5-FU up to 12 h by extending its short original plasma half-life of 15–20 min. The molar mass between cross-links and cross-link density data supported the formation of dense polymer matrix. The diffusional exponent (n) values calculated by Ritger–Peppas equation as well as other empirical equations suggested non-Fickian release of 5-FU.

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